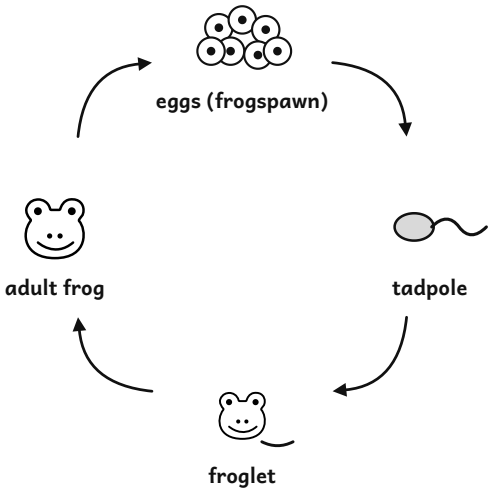
LIVING THINGS

Life Cycles, Growth and Inheritance

Life cycles of plants, chickens, frogs and butterflies, what living things need, inheritance and variation



In this pack

Learn it: Life cycles, growing up and family likeness ★☆☆
Fact check: Life cycle stages ★☆☆
Think it through: Comparing life cycles ★★☆
Investigate: What do seeds need to grow? ★★☆
Real-world case study: Rajshahi silk: the life cycle of the silkworm ★★★
Review and challenge: Show what you know ★★★

This pack belongs to: _____ **Class:** _____

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Life cycles, growing up and family likeness

★☆☆ Grade 4

Name: Class: Date:

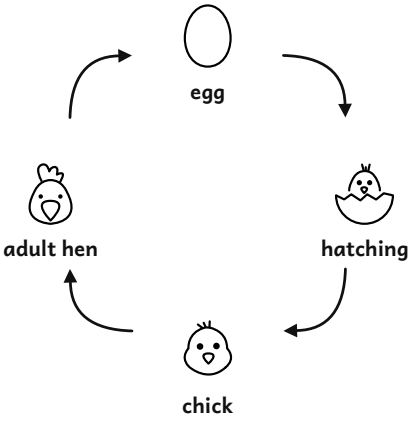
Learning goals

- Describe and compare the life cycles of a plant and some animals.
- Explain what living things need to grow and survive.
- Explain why young living things are like their parents, but not exactly like them.

Every living thing is born, grows, becomes an adult, has young of its own and one day dies. Then its young do the same. This pattern goes round and round, so we call it a **life cycle**. Some life cycles are quick: a mosquito grows from egg to adult in about ten days. Others are slow: an elephant takes about 15 years to become an adult.

A chicken's life cycle

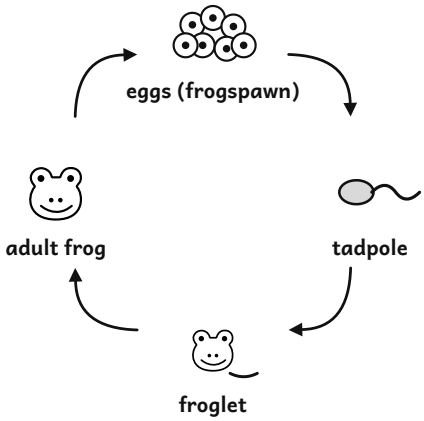
A hen lays an **egg**. Inside, a chick grows for about 21 days, then **hatches**. The fluffy **chick** grows into a young bird, and after about five months it is an **adult** that can lay eggs. Chickens look like small adults from the moment they hatch.



The life cycle of a chicken.

A frog's life cycle

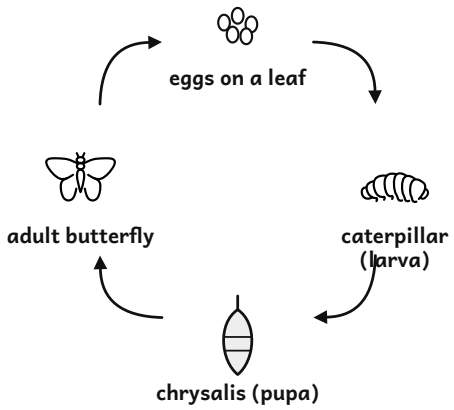
Frogs lay soft **eggs** (frogspawn) in water. A **tadpole** hatches. It has a tail and breathes with **gills**, like a fish. Slowly it grows legs, its tail shrinks and it grows lungs. It becomes a **froglet**, then an **adult frog** that can live on land. This big change of body shape is called **metamorphosis**.



The life cycle of a frog.

A butterfly's life cycle

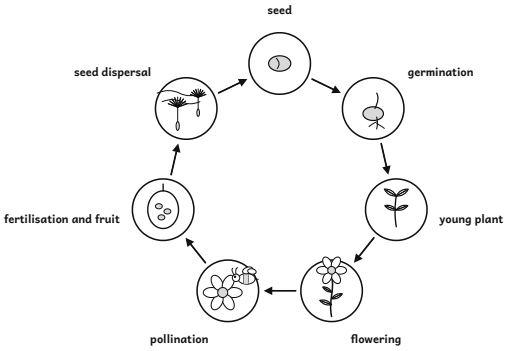
A butterfly lays tiny **eggs** on a leaf. A **caterpillar** (larva) hatches and eats and eats. It then makes a hard case called a **chrysalis** (pupa). Inside, its body changes completely. Finally an **adult butterfly** comes out. This is also metamorphosis.



The life cycle of a butterfly.

A flowering plant's life cycle

A **seed germinates** (starts to grow) into a **seedling**. It grows into an adult plant that makes **flowers**. Pollen is carried from flower to flower (**pollination**), seeds form inside a **fruit**, and the seeds are **scattered** (dispersed). Each seed can grow into a new plant.



The life cycle of a flowering plant.

Compare the life cycles

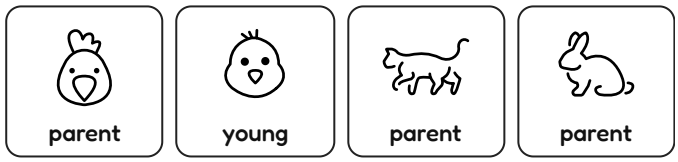
	Plant	Chicken	Frog
starts as	seed	egg	egg
young	seedling	chick	tadpole
big change of shape?	yes	no	yes
makes new young by	seeds	eggs	eggs

What do living things need?

- **Plants** need light, water, air, warmth and minerals from the soil.
- **Animals** need food, water, air (oxygen), shelter and the right temperature.
- If something is missing, a living thing cannot grow well and may not survive.

Like parents, but not exactly

Young living things **inherit** features from their parents: kittens look like cats, mango seeds grow into mango trees, and you may have your mother's eyes or your father's smile. But young are never **exactly** the same as their parents or each other. Kittens in one litter can have different colours and patterns. These small differences are called **variation**.



Remember

Inherited features come from parents (eye colour, the shape of your ears). Some features are not inherited: a scar, short hair after a haircut, or speaking Bangla.

Stay safe

Medicines help our bodies when we are ill. They are **drugs**: substances that change how the body works. Only take medicine given by a trusted adult or a doctor, take the right amount at the right time, and never share it. Keep medicines away from small children.

Key words

life cycle	the stages in the life of a living thing, repeating again and again
hatch	come out of an egg
larva	the young stage of an insect, such as a caterpillar
pupa	the resting stage when an insect changes shape
metamorphosis	a big change of body shape as an animal grows
germinate	when a seed starts to grow
inherit	get features from your parents
variation	differences between living things of the same kind

FACT CHECK · A3 · LIFE CYCLES

Life cycle stages

★☆☆
Grade 4

Name: Class: Date:

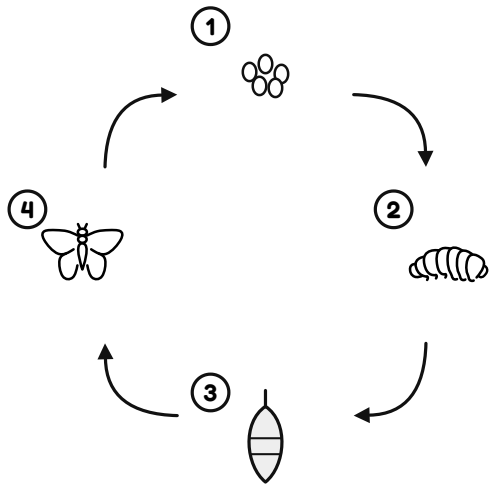
Learning goals

- Order life cycle stages.
- Use life cycle vocabulary.

1 **Number the stages of a frog's life cycle in order.** [4]

- ☐ an adult frog
- ☐ eggs (frogspawn) in the water
- ☐ a froglet with a short tail
- ☐ a tadpole with a tail and gills
- ☐ a tadpole with back legs

2 **Label the butterfly life cycle.** [4]



Number	Stage
1	
2	
3	
4	

3 **Fill in the gaps.** [4]

WORD BANK

metamorphosis **hatches** **inherit** **germinates** **migrates** **melts**

A seed _____ into a seedling. A chick _____ from an egg. A big change of body shape is called _____. Young living things _____ features from their parents.

4 **Sort the features.** [6]

SORT THESE

shape of nose **eye colour** **a scar on your knee** **a haircut** **a puppy's fur colour** **the language you speak**

Inherited from parents	Not inherited

THINK IT THROUGH · A3 · LIFE CYCLES

Comparing life cycles

★★☆
Grade 4

Name: Class: Date:

Learning goals

- Compare life cycles.
- Explain variation.

1 **Give one way the life cycle of a frog is similar to a chicken's and one way it is different.** [2]

2 **Why does a butterfly lay its eggs on a leaf and not on a stone?** [2]

3 **Five kittens are born to the same mother. Two are black, two are striped and one is white. Explain why they are not all the same.** [2]

4 **Tanvir says, "If I grow big muscles at the gym, my children will be born with big muscles." Is he right? Explain.** [2]

5 **Why must you never take medicine that belongs to someone else?** [1]

INVESTIGATE · A3 · LIFE CYCLES

What do seeds need to grow?

★★☆
Grade 4

Name: Class: Date:

Learning goals

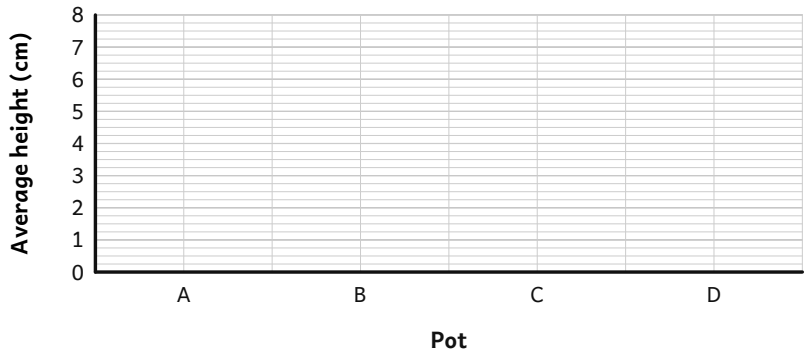
- Plan a fair test.
- Present results in a bar chart.
- Draw conclusions.

Class 4C planted 10 cress seeds in each of four pots. They changed one thing for each pot and measured the height of the seedlings after 7 days.

1 **Here is the plan. What did they change for each pot?** [2]

Pot	Water?	Light?	Warm?	Average height after 7 days (cm)
A	yes	yes	yes	6
B	no	yes	yes	0
C	yes	no (in a dark cupboard)	yes	7 (yellow and thin)
D	yes	yes	no (in a fridge)	1

2 **Draw a bar chart of the heights.** [3]



3 **Pot C was tallest. Does that mean seedlings grow best in the dark? Explain.** [2]

4 **Write a conclusion.** [2]

5 **Name two things they kept the same.** [1]

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LEARN · PLAY · WONDER

Rajshahi silk: the life cycle of the silkworm

★★★ Grade 4

Name: Class: Date:

Learning goals

- Apply life cycle knowledge to a real industry.
- Use numbers from a real process.
- Think about the ethics of using animals.

Science · History · Economics · Ethics · Art

Rajshahi is famous for its beautiful **silk** saris. Silk is made by the **silkworm**, the caterpillar of a moth. Farmers grow **mulberry** trees, because silkworms eat only mulberry leaves.

A silk moth lays 300 to 500 tiny **eggs**. The silkworms (larvae) hatch and eat leaves day and night for about a month, growing about 10 000 times heavier! Then each one spins a **cocoon** around itself from a single thread of silk, which can be **one kilometre long**. Inside the cocoon it becomes a **pupa**. After about two weeks an adult **moth** comes out, and the cycle starts again.

To get long unbroken threads, most cocoons are put in hot water before the moth comes out, which kills the pupa. Some makers produce "peace silk" by waiting for the moths to leave first, but the thread is broken into shorter pieces.

Based on information about sericulture (silk farming) in Rajshahi, Bangladesh.

1 **Complete the silkworm life cycle.** [3]

- ☐ adult moth lays eggs
- ☐ egg
- ☐ pupa inside the cocoon
- ☐ silkworm (larva) eating mulberry leaves
- ☐ spins a cocoon

2 **Why must silk farmers also be good at growing trees?** [1]

.....

.....

3 **Maths. Suppose one sari needs 3000 cocoons. A farmer's silkworms make 12 000 cocoons. How many saris can be made?** [2]

4 **What might each person think about killing the pupae to make silk?** [3]

Person	View
a silk farmer	
an animal-welfare worker	
a sari buyer	

Show what you know

★★★
Grade 4

Name: Class: Date:

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 Find the key words.

R	F	L	S	U	E	U	D	W	D
W	L	A	D	U	L	T	K	G	I
U	S	I	N	H	E	R	I	T	P
I	E	L	Y	M	C	Y	C	L	E
H	E	T	A	D	P	O	L	E	U
U	D	E	E	G	G	F	M	J	Y
R	B	H	A	T	C	H	O	H	G
B	A	I	J	W	M	V	A	L	G
K	P	U	P	A	L	A	R	V	A
J	M	G	M	I	S	B	L	H	P

adult cycle egg hatch inherit larva pupa seed tadpole

2 Which animal goes through metamorphosis? [1]

- ◯ chicken
- ◯ cow
- ◯ butterfly
- ◯ goat

3 What does a seed need to germinate? [1]

- ◯ water and warmth
- ◯ light only
- ◯ soil only
- ◯ wind

4 Draw the life cycle of a plant you know, such as rice or mango. Label four stages. [4]

My plant life cycle

5 **Challenge. Many frogs in Bangladesh are disappearing because ponds and wetlands are being filled in for building. Use the frog life cycle to explain why losing ponds is a problem for frogs.** [2]

.....

.....

.....

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
describe the life cycles of a chicken, a frog, a butterfly and a plant	○	○	○
explain metamorphosis	○	○	○
investigate what seeds need to grow	○	○	○
explain inherited features and variation	○	○	○
use medicines safely	○	○	○
explain how silk comes from the silkworm life cycle	○	○	○